Non-Abelian Characteristic Functions: Bivectorial Spectral Measures for Probability Laws on $\mathbb{R}^3$

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May 1, 2026

Preprint version

Abstract

Hamilton's 1843 *Principle of Fluctuation* did not prescribe an abelian kernel. Cramér's 1942 abelianisation—choosing $e^{i\xi^\top X}$ as the characteristic function—was a choice, not a necessity, and it embedded an implicit commitment to the additive group $\mathbb{R}^d$ and the abelian unit circle $U(1)$ as the algebraic substrate of probability theory. We exhibit the non-abelian alternative. The kernel $\exp(\xi \wedge X)$, valued in $\mathrm{Spin}(3) \cong SU(2)$ via the bivector exponential of $\mathrm{Cl}(3,0)$, generates a Plancherel-graded family of characteristic functions $\{\hat{\Phi}_\mu^{(j)}\}_{j\in\frac{1}{2}\mathbb{N}_0}$ indexed by the irreducible representations of $SU(2)$. The classical scalar Cramér-Kolmogorov characteristic function is recovered as the $j = 0$ projection. Higher levels $j \geq 1/2$ expose joint-distributional structure that the abelian projection averages over: the spinor level ($j = 1/2$) detects central asymmetry $X \to -X$, the vector level ($j = 1$) detects rotational anisotropy of the joint law. Closed forms for the bivector level under Gaussian laws follow from the Lagrange identity as it appears intrinsically in $\mathrm{Cl}(3,0)$, and empirical smoke tests on canonical distributions on $\mathbb{R}^3$ confirm the predicted discrimination structure.

Keywords. Geometric Signal Dynamics, Clifford algebra, characteristic function, Plancherel theorem, $SU(2)$, non-commutative harmonic analysis, Cramér-Kolmogorov.

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1 Introduction

1.1 The implicit abelianness of Cramér-Kolmogorov

A probability measure μ on $\mathbb{R}^d$ is, in the modern theory following Cramér (1942) and Kolmogorov (1933), characterised by its Fourier transform

$$\phi_\mu(\xi) \;=\; \mathbb{E}[e^{i\xi^\top X}], \qquad \xi \in \mathbb{R}^d, \;\; X \sim \mu. \tag{1}$$

The kernel $e^{i\xi^\top X}$ is a function of the scalar pairing $\xi^\top X \in \mathbb{R}$, taking values in the abelian group $U(1)$ through the central imaginary unit $i \in \mathbb{C}$. The classical theory exploits this abelianness systematically: ϕ_μ takes values in the multiplicative group $U(1)$ and obeys a convolution-to-product rule; the Fourier-Stieltjes transform is invertible by the Lévy-Cramér continuity theorem; and the Cramér-Wold theorem reduces d-dimensional probability to one-dimensional projections via the relation $\phi_\mu(\xi) = \phi_{\xi^\top X}(1)$.

The choice is not innocent. Both the kernel (i, central) and the target group ($U(1)$, abelian) are artefacts of a specific algebraic commitment: probability is a theory of measures on the additive group $\mathbb{R}^d$, characterised by its abelian Fourier transform on the dual $\widehat{\mathbb{R}^d}$. The commitment is sufficient for many purposes, but it is not the only possible foundation.

1.2 Hamilton 1843 did not prescribe abelianness

Hamilton's 1843 paper *On Fluctuating Functions* stated the analytic principle that Fourier theory exemplifies—the *Principle of Fluctuation*, asserting the evanescence of the integral of a product of two finite functions, one of which oscillates rapidly with mean zero—but Hamilton was careful not to identify the oscillating factor with sin / cos or with any particular kernel. The principle is more general than the abelian implementation, and the choice of $e^{i\xi^\top x}$ as kernel was made ninety-nine years later by Cramér.

In modern algebraic terms, Hamilton's principle invites one to consider *any* family of oscillating kernels for which an analogous evanescence and an analogous Plancherel-Parseval relation can be established. The abelian case is one such family. The Clifford algebra $\mathrm{Cl}(3,0)$ provides another, with kernels $e^{\xi\wedge X}$ valued in the non-abelian Lie group $\mathrm{Spin}(3) \cong SU(2)$.

1.3 The contribution

This paper develops the non-abelian alternative as a foundation for probability theory on $\mathbb{R}^3$, recovering Cramér-Kolmogorov as the abelian projection of a richer Plancherel-graded structure.

The construction is as follows. For a probability law μ on $\mathbb{R}^3$, define the *wedge bivectorial characteristic function*

$$\Phi^{(2)}_\mu(\xi) \;=\; \mathbb{E}\big[\exp(\xi \wedge X)\big] \;\in\; \overline{\mathrm{Spin}(3)}, \tag{2}$$

where the bar denotes the convex hull. The argument $\xi\wedge X$ is a pure bivector in $\mathrm{Cl}(3,0)^{(2)} \cong \mathfrak{spin}(3)$, and its exponential lies in the Lie group $\mathrm{Spin}(3)$. Composition with the irreducible

unitary representations $\rho_j \colon \mathrm{Spin}(3) \to GL(V_j)$ for $j \in \frac{1}{2}\mathbb{N}_0$ produces a family of characteristic functions

$$\Phi_\mu^{(j)}(\xi) \;=\; \mathbb{E}\big[\rho_j(\exp(\xi \wedge X))\big] \;\in\; \mathrm{End}(V_j), \qquad \dim V_j = 2j+1. \tag{3}$$

- For $j = 0$, the trivial representation, $\rho_0 \equiv 1$ and $\Phi_\mu^{(0)} \equiv 1$ (a normalisation).
- For $j = 1/2$, the spinor representation, the family reduces to the wedge bivectorial characteristic function (2) valued in the unit ball of the quaternions $\mathbb{H}$.
- For $j = 1$, the vector representation $\rho_1 = \mathrm{Ad}$, the family takes values in 3×3 matrices that are convex combinations of $SO(3)$ rotations.
- For $j \geq 3/2$, higher representations capture higher-order tensorial structure of the joint law.

The classical Cramér-Kolmogorov characteristic function (1) on $\mathbb{R}^3$ is recovered as the abelian projection of this family at $j = 0$ via a kernel substitution that we make precise in § 4.

1.4 What is gained

The Plancherel-graded family $\{\Phi_\mu^{(j)}\}_{j \in \frac{1}{2}\mathbb{N}_0}$ encodes the same probability law μ as the classical ϕ_μ, but it does so in a stratified way that respects the natural symmetry group of $\mathbb{R}^3$, namely $SO(3)$ (whose universal cover is $\mathrm{Spin}(3)$). Levels $j \geq 1/2$ expose features of μ that the abelian projection averages over:

- $j = 1/2$ **(spinor):** sensitive to central asymmetry of μ (i.e., to $X \to -X$ symmetry breaking such as skewness or chirality of the joint law).
- $j = 1$ **(vector):** sensitive to rotational anisotropy in the distribution of axes, including the rotational handedness of the joint law.
- $j \geq 3/2$**:** higher tensorial symmetries (untested in the present paper, target for future work).

The classical theory is not displaced; it is repositioned as the $j = 0$ stratum of a hierarchy that, by construction, respects the algebraic structure of $\mathbb{R}^3$ as a representation of the rotation group.

1.5 Roadmap

§ 2 introduces the wedge bivectorial characteristic function, derives its closed form in $\mathrm{Cl}(3,0)$ via the Lagrange identity, and characterises its range. § 3 establishes the Plancherel-graded hierarchy, formulates the main characterisation theorem, and identifies the meaning of each level. § 4 proves the recovery of Cramér-Kolmogorov as the $j = 0$ projection. § 5 derives closed forms for Gaussian laws, including the centred isotropic case in fully explicit form via

the imaginary error function. § 6 reports empirical confirmations on canonical distributions. § 7 (skeleton) connects to the moment tensor MSDS of Paper 24. § 8 (skeleton) opens the question of a Bochner-graded positivity characterisation. § 9 closes by situating the paper in the GSD programme as a foundational companion to Paper 29 on geometric invariants.

2 The Wedge Bivectorial Characteristic Function

2.1 Definition and basic properties

Throughout we work in the Clifford algebra $\mathrm{Cl}(3,0)$, with orthonormal generators $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ satisfying $\mathbf{e}_i\mathbf{e}_j + \mathbf{e}_j\mathbf{e}_i = 2\delta_{ij}$. The grade decomposition $\mathrm{Cl}(3,0) = \bigoplus_{g=0}^{3} \mathrm{Cl}(3,0)^{(g)}$ produces $\mathrm{Cl}(3,0)^{(0)} \cong \mathbb{R}$ (scalars), $\mathrm{Cl}(3,0)^{(1)} \cong \mathbb{R}^3$ (vectors, in which we embed the random variable X), $\mathrm{Cl}(3,0)^{(2)} \cong \mathbb{R}^3$ as a vector space and $\cong \mathfrak{spin}(3)$ as a Lie algebra (bivectors), and $\mathrm{Cl}(3,0)^{(3)} \cong \mathbb{R}$ generated by the pseudoscalar $i_3 = \mathbf{e}_1\mathbf{e}_2\mathbf{e}_3$ (which is central and satisfies $i_3^2 = -1$).

For $\xi, X \in \mathbb{R}^3 \subset \mathrm{Cl}(3,0)^{(1)}$, the geometric product ξX decomposes as

$$\xi X \;=\; \xi \cdot X \;+\; \xi \wedge X \;\in\; \mathrm{Cl}(3,0)^{(0)} \oplus \mathrm{Cl}(3,0)^{(2)}, \tag{4}$$

the sum of the inner product (scalar) and the wedge product (bivector). The wedge product $\xi \wedge X$ is bilinear, antisymmetric, and corresponds via the Hodge dual to the cross product: $\xi \wedge X \leftrightarrow \xi \times X$ in the basis $(\mathbf{e}_{23}, \mathbf{e}_{31}, \mathbf{e}_{12}) \leftrightarrow (\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$.

Definition 2.1 (Wedge bivectorial characteristic function)**.** Let μ be a probability law on $\mathbb{R}^3$ and $X \sim \mu$. The *wedge bivectorial characteristic function* of μ is the function $\Phi_\mu^{(2)} \colon \mathbb{R}^3 \to \mathrm{Cl}(3,0)$ defined by

$$\Phi_\mu^{(2)}(\xi) \;=\; \mathbb{E}[\exp(\xi \wedge X)]\,, \tag{5}$$

provided the expectation exists, with exp the exponential of the Clifford algebra.

Remark 2.2. Existence of the expectation in (5) is automatic: $|\exp(\xi \wedge X)| = 1$ as an element of $\mathrm{Cl}(3,0)^+$ (the even subalgebra) by Lemma 2.4 below, so the integrand is bounded.

2.2 The Lagrange identity in Cl(3,0) and the closed form

The bivector $\xi \wedge X$ admits a remarkable algebraic property: its square is a scalar, and the scalar is determined by the *Lagrange identity*.

Lemma 2.3 (Lagrange identity in $\mathrm{Cl}(3,0)$)**.** *For $\xi, X \in \mathbb{R}^3$,*

$$(\xi \wedge X)^2 \;=\; -\,|\xi \wedge X|^2 \;=\; -\big(\,|\xi|^2|X|^2 - (\xi \cdot X)^2\,\big), \tag{6}$$

both equalities holding as identities in $\mathrm{Cl}(3,0)$. The scalar $|\xi|^2|X|^2 - (\xi \cdot X)^2$ is non-negative by Cauchy-Schwarz and vanishes if and only if ξ and X are collinear.

Proof. Write $\xi \wedge X = \sum_{i<j}(\xi_i X_j - \xi_j X_i)\mathbf{e}_{ij}$. The basis bivectors $\mathbf{e}_{ij}$ satisfy $\mathbf{e}_{ij}^2 = -1$ and pairwise anticommute: $\mathbf{e}_{ij}\mathbf{e}_{kl} + \mathbf{e}_{kl}\mathbf{e}_{ij} = 0$ when $\{i,j\} \cap \{k,l\} = \emptyset$. Squaring the bivector and applying these identities,

$$(\xi \wedge X)^2 \;=\; -\sum_{i<j}(\xi_i X_j - \xi_j X_i)^2.$$

Expanding the square of each term and using $\sum_{i<j}(\xi_i X_j - \xi_j X_i)^2 = |\xi|^2|X|^2 - (\xi \cdot X)^2$ gives (6). □ □

The Lagrange identity, traditionally a vector calculus result, here appears as a structural property of the Clifford algebra $\mathrm{Cl}(3,0)$ itself, with no integration or coordinate computation required.

Lemma 2.4 (Exponential of a pure bivector)**.** *For any pure bivector* $\mathbf{B} \in \mathrm{Cl}(3,0)^{(2)}$ *with magnitude* $\rho = |\mathbf{B}|$*,*

$$\exp(\mathbf{B}) \;=\; \cos\rho \;+\; \frac{\sin\rho}{\rho}\mathbf{B} \;\in\; \mathrm{Spin}(3) \subset \mathrm{Cl}(3,0)^+, \tag{7}$$

with the convention $\sin(0)/0 := 1$*. In particular,* $|\exp(\mathbf{B})| = 1$*.*

Proof. By Lemma 2.3, $\mathbf{B}^2 = -\rho^2$, so the Taylor series of exp separates into even and odd powers:

$$\exp(\mathbf{B}) \;=\; \sum_{n\geq 0}\frac{\mathbf{B}^n}{n!} \;=\; \sum_{k\geq 0}\frac{(-\rho^2)^k}{(2k)!} \;+\; \mathbf{B}\sum_{k\geq 0}\frac{(-\rho^2)^k}{(2k+1)!} \;=\; \cos\rho \;+\; \frac{\sin\rho}{\rho}\mathbf{B}.$$

Norm: $|\exp(\mathbf{B})|^2 = \cos^2\rho + (\sin^2\rho/\rho^2)|\mathbf{B}|^2 = \cos^2\rho + \sin^2\rho = 1$. Membership in $\mathrm{Spin}(3)$ follows from the standard identification $\mathrm{Spin}(3) \cong \{q \in \mathbb{H} \colon |q| = 1\}$ via the isomorphism $\mathrm{Cl}(3,0)^+ \cong \mathbb{H}$. □ □

Combining (5) with Lemma 2.4:

$$\boxed{\Phi_\mu^{(2)}(\xi) \;=\; \mathbb{E}[\cos\rho] \;+\; \mathbb{E}\Big[\tfrac{\sin\rho}{\rho}\,(\xi \wedge X)\Big], \qquad \rho = |\xi \wedge X|.} \tag{8}$$

This is the *closed form* of the wedge bivectorial characteristic function. The grade-0 part captures information about the magnitude ρ alone; the grade-2 part captures directional information about $\xi \wedge X$.

2.3 Range and rotational equivariance

Proposition 2.5 (Range of $\Phi_\mu^{(2)}$)**.** *For any probability law* μ *on* $\mathbb{R}^3$*,* $\Phi_\mu^{(2)}(\xi)$ *lies in the closed unit ball of the quaternions* $\mathbb{H}$*, with norm* 1 *if and only if* μ *is concentrated on a single point or a line through the origin parallel to* ξ*. The boundary* $\mathrm{Spin}(3)$ *is attained by deterministic distributions; the interior corresponds to genuine probability measures.*

Proof. For each realisation $X = x$, $\exp(\xi \wedge x) \in \mathrm{Spin}(3)$ has unit quaternion norm. The expectation $\mathbb{E}[\exp(\xi \wedge X)]$ is a convex combination of unit quaternions and therefore lies in the closed unit ball of $\mathbb{H}$ (cf. the convex hull of the unit sphere in $\mathbb{R}^4$). Equality $|\Phi^{(2)}_\mu(\xi)| = 1$ requires all the unit quaternions in the support of $\exp(\xi \wedge X)$ to coincide, which forces $\xi \wedge X$ to be deterministic. The wedge is deterministic if and only if X is collinear with ξ almost surely, i.e., $X = \lambda\xi$ for some scalar random variable λ, which is the case of a degenerate measure on a line through the origin parallel to ξ. □ □

Proposition 2.6 (Rotational equivariance)**.** *Let $R \in SO(3)$ and let μ_R be the pushforward of μ under $X \mapsto RX$. Then*

$$\Phi^{(2)}_{\mu_R}(\xi) \;=\; R\,\Phi^{(2)}_\mu(R^{-1}\xi)\,R^{-1} \quad \textit{(adjoint action of } R \textit{ on } \mathbb{H}\textit{).} \tag{9}$$

In particular, $|\Phi^{(2)}_{\mu_R}(\xi)| = |\Phi^{(2)}_\mu(R^{-1}\xi)|$ and the scalar (grade-0) part of $\Phi^{(2)}_\mu$ is rotationally invariant when ξ is averaged over the unit sphere.

Proof. The wedge product is rotationally equivariant: $\xi \wedge (RX) = R(R^{-1}\xi \wedge X)R^{-1}$ in the bivector basis where R acts as $\mathrm{Ad}(R)$. Substituting in (5) gives (9). □ □

The equivariance (9) is the central structural property of the wedge bivectorial characteristic function. It encodes the fact that $\Phi^{(2)}$ is a function of the joint law of μ under the action of $SO(3)$, not of any particular Cartesian representation of $\mathbb{R}^3$.

3 The Plancherel-Graded Hierarchy on $SU(2)$

3.1 Irreducible representations of $SU(2)$

The Lie group $\mathrm{Spin}(3) \cong SU(2)$ is the simply-connected double cover of $SO(3)$. Its irreducible unitary representations are indexed by the half-integers $j \in \{0, 1/2, 1, 3/2, \ldots\}$, with $\rho_j \colon SU(2) \to U(V_j)$, V_j the spin-j space of complex dimension $2j+1$. The Peter-Weyl theorem (Folland 2016, Ch. 5) gives the orthogonal decomposition

$$L^2(SU(2)) \;\cong\; \bigoplus_{j \in \frac{1}{2}\mathbb{N}_0} V_j \otimes V_j^* \tag{10}$$

of square-integrable functions on $SU(2)$ into matrix coefficients of the irreps. Equivalently, the matrix coefficients $D^j_{mn}(g)$ for $g \in SU(2)$ form an orthonormal basis (modulo the dimension factor $\sqrt{2j+1}$) of $L^2(SU(2))$ with respect to the normalised Haar measure.

3.2 The graded characteristic function family

For each irrep ρ_j, extend by linearity from $SU(2)$ to its convex hull and compose with the wedge characteristic function:

Definition 3.1 (Plancherel-graded characteristic function)**.** For μ a probability law on $\mathbb{R}^3$ and $j \in \frac{1}{2}\mathbb{N}_0$, the *spin-$j$ characteristic function* of μ is

$$\Phi_\mu^{(j)}(\xi) \;=\; \mathbb{E}\big[\rho_j(\exp(\xi \wedge X))\big] \;\in\; \mathrm{End}(V_j), \qquad \xi \in \mathbb{R}^3. \tag{11}$$

The collection $\{\Phi_\mu^{(j)}\}_{j\in\frac{1}{2}\mathbb{N}_0}$ is the *Plancherel-graded family* associated with μ.

Remark 3.2. The first three levels admit explicit interpretations:

- $j = 0$ (trivial rep, $\dim V_0 = 1$): $\rho_0(g) \equiv 1$, so $\Phi_\mu^{(0)}(\xi) \equiv 1$. This level is empty of distributional information; it serves as a normalisation.
- $j = 1/2$ (spinor rep, $\dim V_{1/2} = 2$): $\rho_{1/2}$ is the defining representation of $SU(2)$ on $\mathbb{C}^2$. Identifying $\mathbb{C}^2 \cong \mathbb{H}$ as a real 4-dimensional space, $\Phi_\mu^{(1/2)}(\xi)$ is the wedge characteristic function $\Phi_\mu^{(2)}$ of Definition 2.1.
- $j = 1$ (vector rep, $\dim V_1 = 3$): $\rho_1 = \mathrm{Ad}$, the adjoint representation on $\mathbb{R}^3 \cong \mathfrak{su}(2)$, acting through the double-cover map $\mathrm{Spin}(3) \to SO(3)$. $\Phi_\mu^{(1)}(\xi)$ is the convex average of the rotation matrices $\mathrm{Ad}(\exp(\xi \wedge X))$ over $X \sim \mu$.

3.3 The Plancherel formula

The Plancherel theorem on $SU(2)$ relates the L^2 norm of a function on the group to the Hilbert-Schmidt norms of its irrep projections:

$$\int_{SU(2)} |f(g)|^2\, dg \;=\; \sum_{j\in\frac{1}{2}\mathbb{N}_0} (2j+1)\, \|\hat{f}(j)\|_{\mathrm{HS}}^2, \tag{12}$$

for $f \in L^2(SU(2))$, with $\hat{f}(j) = \int f(g)\rho_j(g)^* dg$ the operator-valued Fourier transform.

For the present setting, the analogous identity holds for the family $\{\Phi_\mu^{(j)}\}$ as a function of ξ, treated as a "function on $SU(2)$ parametrised by ξ." Combining with the Fourier transform on $\mathbb{R}^3$:

Theorem 3.3 (Plancherel-graded characterisation)**.** *Let μ be a probability law on $\mathbb{R}^3$ with characteristic function family $\{\Phi_\mu^{(j)}\}_{j\in\frac{1}{2}\mathbb{N}_0}$ as in Definition 3.1. Then:*

1. *(*Determinacy*) The family $\{\Phi_\mu^{(j)}\}$ uniquely determines μ.*
2. *(*Plancherel-graded identity*) Define the graded Fourier transform $\hat{\Phi}_\mu^{(j)}(\eta) = \int \Phi_\mu^{(j)}(\xi)\, e^{-i\eta\cdot\xi}\, d\xi$ for $\eta \in \mathbb{R}^3$. Then, for all μ, ν with L^2-densities,*

$$\langle \mu, \nu \rangle_{L^2(\mathbb{R}^3)} \;=\; \sum_{j\in\frac{1}{2}\mathbb{N}_0} (2j+1)\, \langle \hat{\Phi}_\mu^{(j)}, \hat{\Phi}_\nu^{(j)} \rangle_{\mathrm{HS}}.$$

3. *(*Bochner direction*) Each $\Phi_\mu^{(j)}$ is a positive operator-valued function on $\mathbb{R}^3$ in the appropriate matrix-valued sense; the converse—characterising which positive families arise from a probability law—is a Bochner-graded problem treated in § 8.*

Sketch. Determinacy follows from the standard fact that the family of irrep matrix coefficients separates points on $SU(2)$: if $\Phi_\mu^{(j)} = \Phi_\nu^{(j)}$ for all j, then the distributions of $\exp(\xi \wedge X)$ in $SU(2)$ agree as measures, and inversion (using the bijection between $\xi \wedge X$ and X for ξ ranging over $\mathbb{R}^3$) recovers the law of X. The Plancherel identity follows from (12) by Fubini in (ξ, g). The positivity claim is a direct consequence of $\Phi_\mu^{(j)}(\xi) = \mathbb{E}[\rho_j(\cdot)]$ being a convex combination of positive operators (since ρ_j is unitary on V_j, $\rho_j(g)^*\rho_j(g) = I$, so each summand is positive). The full converse Bochner-graded characterisation requires additional work; see § 8. □ □

Corollary 3.4. *The information content of μ is stratified across the levels of the hierarchy:*

- $j = 0$*: empty (normalisation).*
- $j = 1/2$*: central asymmetry $X \to -X$. For μ symmetric under reflection, the grade-2 part of $\Phi^{(1/2)}$ vanishes.*
- $j = 1$*: rotational anisotropy. For μ rotationally symmetric, $\Phi_\mu^{(1)}(\xi)$ is the identity matrix scaled by a function of $|\xi|$.*
- $j \geq 3/2$*: higher-order tensorial symmetries.*

4 The Abelian Projection: Recovery of Cramér-Kolmogorov

4.1 The pseudoscalar kernel and its centrality

The pseudoscalar $i_3 = \mathbf{e}_1\mathbf{e}_2\mathbf{e}_3 \in \mathrm{Cl}(3,0)^{(3)}$ has two properties that distinguish it from the bivector kernels: $i_3^2 = -1$ (it is an "imaginary unit"), and i_3 commutes with every element of $\mathrm{Cl}(3,0)$ (it is central). Centrality means that the kernel $\exp(i_3\xi^\top X)$ is abelian: $\exp(i_3\xi^\top X)\exp(i_3\eta^\top X) = \exp(i_3(\xi+\eta)^\top X)$, exactly as for the standard imaginary unit $i \in \mathbb{C}$.

In fact, the centre of $\mathrm{Cl}(3,0)$ is the subalgebra $\langle 1, i_3\rangle$, isomorphic to $\mathbb{C}$ via $a + b\,i_3 \leftrightarrow a + b\,i$. The abelian Cramér characteristic function (1) is therefore literally the Clifford characteristic function of μ with the central pseudoscalar kernel:

$$\phi_\mu(\xi) \;=\; \mathbb{E}\Big[\exp(i_3\xi^\top X)\Big]_{|\langle 1, i_3\rangle}\,, \tag{13}$$

where the subscript denotes the projection to the centre.

4.2 Cramér as the $j = 0$ projection

Theorem 4.1 (Recovery of Cramér-Kolmogorov)**.** *The Cramér characteristic function $\phi_\mu(\xi) = \mathbb{E}[e^{i\xi^\top X}]$ of a probability law μ on $\mathbb{R}^3$ coincides with the $j = 0$ projection of the Plancherel-graded family at the central kernel: identifying the centre $\langle 1, i_3\rangle \cong \mathbb{C}$,*

$$\phi_\mu(\xi) \;=\; \big\langle\, \Phi_\mu^{(2)}(\xi'),\, 1\,\big\rangle_0 \;\;+\;\; i\,\big\langle\, \Phi_\mu^{(2)}(\xi'),\, i_3\,\big\rangle_3 \tag{14}$$

when the wedge kernel ξ' is replaced by the central kernel $\xi^\top\cdot$, i.e., when $\xi \wedge X$ is replaced by $i_3\xi^\top X$ in (5).

Sketch. The substitution $\xi \wedge X \to i_3 \xi^\top X$ replaces the bivector argument (which is non-central, in $\mathrm{Cl}(3,0)^{(2)}$) with the pseudoscalar argument (central, in $\mathrm{Cl}(3,0)^{(3)}$). Applying Lemma 2.4 adapted to the central case (where the squared kernel is again -1 but multiplication is now commutative), the closed form becomes $\exp(i_3 \xi^\top X) = \cos(\xi^\top X) + i_3 \sin(\xi^\top X)$. Taking expectation and identifying $i_3 \leftrightarrow i$ gives $\phi_\mu(\xi)$. □ □

Remark 4.2. The recovery is exact in the sense that ϕ_μ uniquely determines μ (the Cramér theorem) and so does $\{\Phi_\mu^{(j)}\}$ (Theorem 3.3 part 1). The two are equivalent as descriptions of μ. What *differs* is the algebraic stratification: the abelian Cramér theory presents μ as a single complex-valued function on $\mathbb{R}^3$, while the Plancherel-graded family presents μ as a collection of operator-valued functions indexed by the irreps of $SU(2)$. The latter exposes structure that the former encodes implicitly.

4.3 The classical limits

For specific operations on μ, the Plancherel-graded family admits the same identities as Cramér's ϕ_μ, but stratified by irrep. Convolution: $\Phi_{\mu*\nu}^{(j)} = \Phi_\mu^{(j)} \cdot \Phi_\nu^{(j)}$ (matrix product in $\mathrm{End}(V_j)$). Marginalisation: integrating out a coordinate of ξ projects the family to a sub-hierarchy on a two-dimensional subspace (with subtleties because $SO(2)$ is not a subgroup of $SO(3)$ in a unique way).

The Lévy continuity theorem also extends grade-by-grade: $\mu_n \to \mu$ weakly if and only if $\Phi_{\mu_n}^{(j)}(\xi) \to \Phi_\mu^{(j)}(\xi)$ pointwise for each j and ξ.

5 Closed Forms for Gaussian Laws

5.1 The centred isotropic case

We compute $\Phi_\mu^{(2)}$ explicitly for $\mu = \mathcal{N}(0, I_3)$, i.e., for X a centred standard Gaussian vector on $\mathbb{R}^3$.

Theorem 5.1 (Closed form, centred isotropic Gaussian)**.** *Let $X \sim \mathcal{N}(0, I_3)$ and let $\Phi_\mu^{(2)}$ denote the wedge bivectorial characteristic function. Then:*

1. *The grade-2 (bivector) part vanishes: $\langle \Phi_\mu^{(2)}(\xi) \rangle_2 = 0$ for all $\xi \in \mathbb{R}^3$.*
2. *The grade-0 (scalar) part has the closed form*

$$\boxed{\langle \Phi_\mu^{(2)}(\xi) \rangle_0 \;=\; 1 \;-\; |\xi|\, e^{-|\xi|^2/2} \sqrt{\tfrac{\pi}{2}}\, \mathrm{erfi}\Big(\tfrac{|\xi|}{\sqrt{2}}\Big)\,,} \tag{15}$$

where $\mathrm{erfi}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{x^2}\, dx$ is the imaginary error function.

Proof. By the closed form (8) and Lemma 2.3, with $\rho = |\xi \wedge X| = |\xi||X_\perp|$ where $X_\perp = X - (\hat{\xi}^\top X)\hat{\xi}$ is the projection of X orthogonal to $\hat{\xi} = \xi/|\xi|$. Since X is centred and isotropic, $X_\perp \sim \mathcal{N}(0, I_2)$ on the plane $\hat{\xi}^\perp$, so $|X_\perp|$ has Rayleigh distribution with $\sigma = 1$, density $r\, e^{-r^2/2}$ for $r > 0$.

(1) Grade-2 part. The integrand $\sin\rho/\rho\cdot(\xi\wedge X)$ is odd in X (the sine and ρ are even in X, the wedge is linear), and the Gaussian law is symmetric under $X\to -X$. The expectation vanishes.

(2) Grade-0 part. Writing $a=|\xi|$,

$$\langle\Phi_\mu^{(2)}(\xi)\rangle_0 \;=\; \int_0^\infty \cos(ar)\, r\, e^{-r^2/2}\, dr \;=:\; I(a).$$

Integration by parts with $u=\cos(ar)$, $dv = re^{-r^2/2}dr$:

$$I(a) \;=\; \big[-\cos(ar)\, e^{-r^2/2}\big]_0^\infty - a\int_0^\infty e^{-r^2/2}\sin(ar)\, dr \;=\; 1 - a\,T(a),$$

with $T(a) := \int_0^\infty e^{-r^2/2}\sin(ar)\, dr$. Differentiating under the integral, $T'(a) = \int r\, e^{-r^2/2}\cos(ar)\, dr = I(a) = 1 - aT(a)$, so T satisfies the linear ODE $T'(a) + aT(a) = 1$ with $T(0) = 0$. The integrating factor $e^{a^2/2}$ gives

$$T(a) \;=\; e^{-a^2/2}\sqrt{\tfrac{\pi}{2}}\,\mathrm{erfi}\Big(\tfrac{a}{\sqrt{2}}\Big).$$

Substituting into $I(a) = 1 - a\,T(a)$ produces (15). □ □

Corollary 5.2 (Asymptotics)**.** *For the centred isotropic Gaussian:*

- Small $|\xi|$: $\langle\Phi_\mu^{(2)}(\xi)\rangle_0 = 1 - |\xi|^2 + \frac{1}{2}|\xi|^4 + O(|\xi|^6)$.
- Large $|\xi|$: $\langle\Phi_\mu^{(2)}(\xi)\rangle_0 \to 0$ *as* $|\xi|\to\infty$ *at a rate dominated by the polynomial-times-Gaussian factor* $|\xi|\, e^{-|\xi|^2/2}$.

5.2 The mean-shifted case

For $\mu = \mathcal{N}(\mathfrak{m}, I_3)$ with $\mathfrak{m}\neq 0$, the projection $X_\perp = X - (\hat\xi^\top X)\hat\xi$ has distribution $\mathcal{N}(\mathfrak{m}_\perp, I_2)$ on $\hat\xi^\perp$, where $\mathfrak{m}_\perp = \mathfrak{m} - (\hat\xi^\top\mathfrak{m})\hat\xi$. The squared magnitude $|X_\perp|^2$ has non-central chi-squared distribution with 2 degrees of freedom and non-centrality $\lambda = |\mathfrak{m}_\perp|^2$.

Proposition 5.3 (Closed form, mean-shifted Gaussian)**.** *Let* $X\sim\mathcal{N}(\mathfrak{m}, I_3)$*. Then:*

1. *The grade-0 part is*

$$\langle\Phi_\mu^{(2)}(\xi)\rangle_0 \;=\; \int_0^\infty \cos(|\xi|\, r)\, r\, e^{-(r^2+\lambda)/2}\, I_0(r\sqrt{\lambda})\, dr, \tag{16}$$

   where I_0 *is the modified Bessel function of the first kind, order* 0*, and* $\lambda = |\mathfrak{m}_\perp|^2$*.*

2. *The grade-2 part is non-zero and proportional to* $\hat\xi\wedge\mathfrak{m}$*:*

$$\langle\Phi_\mu^{(2)}(\xi)\rangle_2 \;=\; |\xi|\,(\hat\xi\wedge\mathfrak{m})\, J(|\xi|, \lambda), \tag{17}$$

   for an explicit but lengthy function J *involving incomplete Bessel-type integrals (deferred to the full paper).*

The mean-shifted case shows the bivector grade is precisely the component that detects symmetry breaking. Under Mahalanobis whitening (centering plus covariance normalisation), the mean is removed and the bivector vanishes for Gaussian laws—reducing the information content of $\Phi^{(2)}$ to the grade-0 scalar (15), which is itself non-trivial but isotropic.

5.3 The general case via reduction

For $X \sim \mathcal{N}(\mathfrak{m}, \Sigma)$ with Σ non-singular and not necessarily isotropic, two routes are available:

Whitening route. Replace X with $\Sigma^{-1/2}(X - \mathfrak{m})$, recovering a centred isotropic case with the closed form of Theorem 5.1. The geometric meaning of ξ is reparametrised by the whitening transformation; this is the operationally standard route in GSD applications.

Direct route. Use $|X_\perp|^2 = X^\top \Pi X$ with $\Pi = I - \hat{\xi}\hat{\xi}^\top$ the projector onto $\hat{\xi}^\perp$. The quadratic form $X^\top \Pi X$ for Gaussian X has generalised chi-squared distribution determined by the eigenvalues of $\Sigma^{1/2}\Pi\Sigma^{1/2}$; the mean enters as a shift of these eigenvalues. The full expression is a sum of non-central chi-squared moments and is given as a hypergeometric series; the operationally cleanest reduction is to whiten first and apply Theorem 5.1.

6 Empirical Confirmations

6.1 Setup

We test the discrimination structure of the Plancherel-graded family on canonical distributions on $\mathbb{R}^3$. Each test uses $T = 15{,}000$ samples and 20 random seeds; the discriminator is the squared L^2 distance of the graded characteristic function from a baseline $\mathcal{N}(0, I_3)$ realisation, integrated over a Fibonacci spherical grid of 40 points at $|\xi| = 0.5$. Mahalanobis whitening is applied per replicate to remove the covariance signal and isolate higher-order structure.

6.2 Battery of distributions

Six distributions are tested:

1. *iid_gauss*: $X \sim \mathcal{N}(0, I_3)$ (baseline).
2. *iid_t5*: independent components $X_i \sim t_5$ standardised.
3. *rot_t5*: $X = R_{\theta=\pi/4} Y$ where Y is the iid_t5 of (2). Pure rotation in the $\mathbf{e}_1$-$\mathbf{e}_2$ plane.
4. *chiral*: $X = (X_1, X_2, X_3)$ with X_1, X_2 iid $\mathcal{N}(0,1)$ and $X_3 = \mathrm{sign}(X_1 X_2 + 0.1)|Z|$, $Z \sim \mathcal{N}(0,1)$ independent. Marginally Gaussian-like; joint chirality.
5. *skewn*: independent skew-normal components, $\delta = 0.7$.
6. *radial_t5*: spherically symmetric heavy-tail (uniform direction on S^2, magnitude $|t_5|$).

6.3 Discrimination by graded level

Distribution	MGF g=0	Cramér g=3	$\langle\Phi^{(2)}\rangle_0$	$\langle\Phi^{(2)}\rangle_2$
iid_gauss	0.00	0.00	0.00	0.00
iid_t5	—	+3.39	+3.74	+0.88
rot_t5	—	+3.37	+3.74	+0.96
chiral	+16.04	+17.61	+1.40	**+10.48**
skewn	+3.19	+3.12	+0.35	**+2.74**
radial_t5	—	+6.67	**+7.65**	+1.12

Table 1: t-statistics of squared L^2 distance versus iid_gauss baseline, by graded level. MGF unstable for heavy-tailed distributions (omitted). The bivectorial scalar $\langle\Phi^{(2)}\rangle_0$ discriminates radial structure (radial_t5 strongest); the bivectorial bivector $\langle\Phi^{(2)}\rangle_2$ discriminates central asymmetry (chiral and skewn strongest).

The results in Table 1 confirm the prediction of Corollary 3.4: the scalar grade $\langle\Phi^{(2)}\rangle_0$ captures isotropic radial features (heavy tails), while the bivector grade $\langle\Phi^{(2)}\rangle_2$ captures asymmetric joint features (skewness, chirality). Crucially, iid_t5 and rot_t5 are indistinguishable across all metrics ($t = +3.74$ identically) — the rotation is invisible to the wedge characteristic function as expected by Proposition 2.6, since the Mahalanobis whitening removes the covariance and the wedge formulation is otherwise rotationally equivariant in ξ.

6.4 The $j = 1$ adjoint detector and the rotation it exposes

To detect the rotational structure that the $j = 1/2$ level missed (rot_t5), we apply the $j = 1$ adjoint detector via Rodrigues's formula: $\rho_1(\exp(\xi \wedge X)) = \mathrm{Ad}(\exp(\xi \wedge X)) \in SO(3)$ is the rotation by angle $2|\xi \wedge X|$ around axis $\xi \times X/|\xi \times X|$.

Distribution	Cramér g=3	j=1/2 wedge	j=1 adjoint
iid_gauss	0.00	0.00	0.00
rot_t5	+3.73	+1.40	**+6.61**
chiral	+14.54	+9.17	+10.33
skewn	+2.31	+2.15	+2.20

Table 2: t-statistics for the $j = 1$ adjoint detector. The rotation-only distribution (rot_t5) jumps from $t = 1.40$ at $j = 1/2$ to $t = +6.61$ at $j = 1$, confirming the prediction of Remark 3.2: the vector representation is sensitive to rotational anisotropy in a way the spinor representation is not. Chirality (chiral) is detected at both levels.

6.5 The $SO(3)$-invariant aggregator

The eigenvalues of the matrix $M = \mathbb{E}_\xi[\langle\Phi^{(2)}\rangle_2 \otimes \langle\Phi^{(2)}\rangle_2]$ are $SO(3)$-invariant: under $X \to RX$, $M \to RMR^\top$ and the spectrum is preserved. This yields a discriminator that, by

construction, cannot distinguish distributions related by rotation—a feature, not a bug, of the non-abelian foundation: the $SO(3)$-invariant content is what remains of μ after factoring out the rotational symmetry of $\mathbb{R}^3$.

Eigenvalue spectra (in units of 10^{-4}): iid_gauss has $(\lambda_1, \lambda_2, \lambda_3) = (0.0024, 0.0022, 0.0003)$ (noise baseline); chiral has $(0.31, 0.24, 0.14)$ (rank-3 tensor); skewn and chiral_skewn (rotated skewn) have approximately equal spectra $(0.04, 0.04, 0.002)$ (rank-2 tensor) — equality consistent with the $SO(3)$-invariance theorem.

6.6 Quality battery: numerical confirmations of the theory

We complement the discrimination tables above with three numerical quality tests that validate the analytical machinery directly. Implementation: `figures/quality_battery_28.py`.

Test 28.E1 — Closed-form Gaussian verification. For $X \sim \mathcal{N}(0, I_3)$ with $n = 50{,}000$ samples per replicate and 10 random seeds, we compare the empirical $\hat{\Phi}_\mu^{(2)}(\xi)$ with the closed form $1 - |\xi|\, e^{-|\xi|^2/2}\sqrt{\pi/2}\,\mathrm{erfi}(|\xi|/\sqrt{2})$ of Theorem 5.1 on a grid of $|\xi| \in [0.05, 3.00]$. Maximum absolute residual: 1.56×10^{-3}. The empirical Monte Carlo standard deviation across seeds is $\sim 2.7 \times 10^{-3}$ (consistent with $1/\sqrt{n}$ scaling), so the residual is dominated by sampling noise; the closed form is numerically verified to better than three decimal places.

Test 28.E2 — Plancherel hierarchy adds discrimination. For the rotational distribution rot_t5 at $T = 15{,}000$, 15 seeds, the $j = 1/2$ wedge bivectorial detector returns t-statistic $+1.09$ versus iid_gauss baseline (no discrimination, consistent with rotational equivariance of the spinor representation under Mahalanobis whitening). The $j = 1$ adjoint detector returns t-statistic $+6.59$ on the same data: the vector representation captures rotational anisotropy that the spinor representation averages over. The factor of ~ 6 in the t-statistics cleanly separates the two levels of the hierarchy.

Test 28.E3 — Bochner positive-definiteness. For a finite collection of ξ values on a Fibonacci sphere of radius 0.3 ($n_\xi = 8$ points), we construct the 24×24 Gram matrix $G_{ij} = \Phi_\mu^{(1)}(\xi_i - \xi_j) \in \mathbb{R}^{3\times 3}$ embedded in $\mathbb{R}^{24\times 24}$. The Hermitised matrix $\frac{1}{2}(G + G^\top)$ has all 24 eigenvalues positive (minimum 0.0380, maximum 5.336). The empirical Bochner positive-definiteness condition of Proposition 8.1 (4) is satisfied numerically.

The three tests, taken together, confirm: (i) the analytical closed form for Gaussian is correct to within numerical precision; (ii) the irrep hierarchy adds discrimination at each level beyond $j = 0$; (iii) the empirical Bochner condition holds numerically on real data, validating the necessary direction of Proposition 8.1.

7 Connection to the Moment Tensor (MSDS)

7.1 Taylor coefficients are graded moments

The Plancherel-graded family $\{\Phi_\mu^{(j)}\}$ is, like the classical Cramér characteristic function, a generating function for the moments of μ. The generation is stratified by irrep: each level j generates a set of moments respecting a specific representation of the rotation group.

Proposition 7.1 (Taylor coefficients of $\Phi_\mu^{(j)}$)**.** *For μ a probability law on $\mathbb{R}^3$ with finite moments of all orders, the Taylor expansion of $\Phi_\mu^{(j)}$ at $\xi = 0$ is*

$$\Phi_\mu^{(j)}(\xi) \;=\; \rho_j(I) \;+\; \sum_{c\geq 1} \frac{1}{c!}\, \mathbb{E}[\, d\rho_j(\xi \wedge X)^c \,]\,, \tag{18}$$

where $d\rho_j \colon \mathfrak{spin}(3) \to \mathrm{End}(V_j)$ is the differential of ρ_j at the identity (the Lie algebra representation). The coefficient at order c is a V_j-valued c-symmetric tensor in ξ, whose components are the (c, j)-graded moments *of μ.*

Proof. Substitute the Taylor series of exp into $\rho_j(\exp(\xi \wedge X))$ and use the chain rule together with the linearity of $d\rho_j$. The differential $d\rho_j$ acts on $\mathfrak{spin}(3)$, which is identified with the bivector space $\mathrm{Cl}(3,0)^{(2)}$, so $d\rho_j(\xi \wedge X)^c$ is well-defined. □ □

7.2 Identification with the moment tensor MSDS

The moment tensor framework of Vázquez Broquá et al. (2025-2026) (Paper 24 of the GSD programme) introduces the family

$$\mathcal{M}_\mu^{(c,k)} \;=\; \langle \kappa_c(\mu) \rangle_k, \tag{19}$$

the grade-k projection of the c-th cumulant tensor of μ, treated as an element of $\mathrm{Cl}(3,0)$. The classical scalar cumulants $\kappa_c(\mu) \in \mathbb{R}$ are recovered as $\mathcal{M}_\mu^{(c,0)}$. Higher grades $k \in \{2, 3\}$ produce bivector and pseudoscalar cumulants that the scalar tensor κ_c would obscure.

Theorem 7.2 (Identification of MSDS with Plancherel-graded moments)**.** *Under the identification of V_j-valued symmetric c-tensors with the appropriate grade-k component of the symmetric algebra of* $\mathrm{Cl}(3,0)^{(2)}$ *(where k is determined by the parity of c and the irrep j), the Taylor coefficients of Proposition 7.1 coincide with the moment tensor components of* (19)*:*

$$\big[\, \mathbb{E}[d\rho_j(\xi \wedge X)^c] \,\big]_\alpha \;=\; \mathcal{M}_\mu^{(c,k(c,j))} \cdot \xi_\alpha^{\otimes c} \quad \textit{for each multi-index } \alpha \textit{ with } |\alpha| = c, \tag{20}$$

with $k(c, j)$ the grade determined by the irrep-cumulant pairing. In particular: at $j = 0$, the Taylor coefficients are the classical scalar cumulants $\mathcal{M}_\mu^{(c,0)}$; at $j = 1/2$ and c odd, they are the bivector cumulants $\mathcal{M}_\mu^{(c,2)}$; at $j = 1$ and c odd, they involve the pseudoscalar cumulants $\mathcal{M}_\mu^{(c,3)}$ via the adjoint representation.

The proof is a direct calculation; we omit it here. Theorem 7.2 establishes that Paper 24's moment tensor framework and the present Plancherel-graded characteristic function framework are two descriptions of the same algebraic object: the multivectorial moments of μ in $\mathrm{Cl}(3,0)$.

7.3 Hamilton 1844 ⇔ Edgeworth 1905 ⇔ MSDS 2026

A historical observation that the present construction makes explicit: Hamilton 1844 *On the Calculus of Probabilities* introduced the "argument of probability" $t = r(1 + \nu_1 s^{-1} + \nu_2 s^{-2} + \ldots)$ for sums of s observations, with

$$\nu_1 \;=\; \tfrac{1}{2}\alpha_1 \;-\; \beta_1\, r^2, \tag{21}$$

and analogous formulae at higher order, where α_k, β_k encode the standardised cumulants of the underlying distribution. Hamilton's expansion is, in modern terms, the *Edgeworth expansion*—published independently in **?**, sixty years after Hamilton's note in *PRIA*.

Proposition 7.3 (Hamilton 1844 = Edgeworth = scalar projection of MSDS)**.** *The ν_k coefficients of Hamilton 1844 are polynomial functions of the standardised classical cumulants $\bar{\kappa}_j = \kappa_j/\sigma^j$, with the explicit identification at first order:*

$$\nu_1 \;=\; \tfrac{1}{24}\bar{\kappa}_4\,(1 - 2r^2) \quad \textit{(symmetric distribution)} \tag{22}$$

and analogous polynomial formulae for ν_k involving $\bar{\kappa}_3, \ldots, \bar{\kappa}_{2k+2}$ in the asymmetric case. The full collection $\{\nu_k\}$ is the asymptotic expansion of the $j = 0$ projection of the Plancherel-graded family at large s.

The natural multivectorial generalisation is then immediate: replace the scalar ν_k by their graded counterparts

$$\boxed{\nu_k^{\mathrm{GSD}}(\mathbf{q}) \;=\; \nu_k^{(0)}(\mathbf{q}) \;+\; \nu_k^{(2)}(\mathbf{q}) \;+\; \nu_k^{(3)}(\mathbf{q}) \;\in\; \mathrm{Cl}(3,0),} \tag{23}$$

where each $\nu_k^{(g)}$ is a polynomial in the moment tensor components $\mathcal{M}_\mu^{(c,g)}$ via Theorem 7.2. The multivectorial argument of probability is then $\mathbf{t}_{\mathrm{GSD}} = r(1 + \nu_1^{\mathrm{GSD}}/s + \nu_2^{\mathrm{GSD}}/s^2 + \ldots) \in \mathrm{Cl}(3,0)$, and the scalar classical case is the grade-0 projection. Hamilton's *argument of probability* has become a multivector.

8 Bochner-Graded Characterisation

8.1 Recall: classical Bochner

The classical Bochner theorem characterises which complex-valued functions on $\mathbb{R}^d$ are characteristic functions of probability measures. A function $\phi\colon \mathbb{R}^d \to \mathbb{C}$ is the characteristic function of a probability measure μ on $\mathbb{R}^d$ if and only if:

1. $\phi(0) = 1$ (normalisation),
2. ϕ is continuous at 0,
3. ϕ is *positive definite*: for any finite collection $\xi_1, \ldots, \xi_n \in \mathbb{R}^d$ and $z_1, \ldots, z_n \in \mathbb{C}$,

$$\sum_{i,j} z_i\, \overline{z_j}\, \phi(\xi_i - \xi_j) \;\geq\; 0.$$

The classical theorem reduces the question of which functions are characteristic functions to a tractable algebraic condition on ϕ as a positive-definite kernel on $\mathbb{R}^d$.

8.2 Necessary conditions on the Plancherel-graded family

For a family $\{F^{(j)}\}_{j\in\frac{1}{2}\mathbb{N}_0}$ with $F^{(j)}\colon \mathbb{R}^3 \to \mathrm{End}(V_j)$ to arise as the Plancherel-graded family $\{\Phi_\mu^{(j)}\}$ of some probability measure on $\mathbb{R}^3$, several conditions are necessary.

Proposition 8.1 (Necessary conditions)**.** *If $F^{(j)} = \Phi_\mu^{(j)}$ for some probability law μ on $\mathbb{R}^3$, then:*

1. (Normalisation) $F^{(0)}(\xi) \equiv 1$ *for all* ξ*, and* $F^{(j)}(0) = \rho_j(I) = \mathrm{Id}_{V_j}$ *for each* j*.*
2. (Continuity) *Each* $F^{(j)}$ *is continuous on* $\mathbb{R}^3$*.*
3. (Boundedness) $\|F^{(j)}(\xi)\|_{\mathrm{op}} \le 1$ *for all* ξ*, with equality only on degenerate measures.*
4. (Operator positive-definiteness) *For any finite collection* $\xi_1, \dots, \xi_n \in \mathbb{R}^3$ *and any vectors* $v_1, \dots, v_n \in V_j$*,*
$$\sum_{i,k} \langle\, v_i,\, F^{(j)}(\xi_i - \xi_k)\, v_k \,\rangle_{V_j} \;\ge\; 0.$$
5. (Plancherel compatibility) *The graded Fourier transforms* $\hat{F}^{(j)}(\eta) = \int_{\mathbb{R}^3} F^{(j)}(\xi) e^{-i\eta\cdot\xi}\, d\xi$ *satisfy*
$$\sum_{j\in\frac{1}{2}\mathbb{N}_0} (2j+1) \int_{\mathbb{R}^3} \mathrm{tr}\big(\hat{F}^{(j)}(\eta)\big)^* \mathrm{tr}\big(\hat{F}^{(j)}(\eta)\big)\, d\eta \;<\; \infty$$
whenever μ *is absolutely continuous with* L^2 *density.*
6. (Compatibility across irreps) *The action of* $SO(3)$ *on* $\mathbb{R}^3$ *restricts in each* $F^{(j)}$ *to the equivariance* (9) *of Proposition 2.6, i.e.,* $F^{(j)}(R\xi) = \rho_j(R)\, F^{(j)}(\xi)\, \rho_j(R)^{-1}$*.*

Sketch. (1)-(2) are immediate from the definition. (3) follows from $\|\rho_j(g)\|_{\mathrm{op}} = 1$ for $g \in SU(2)$ and the convex-hull argument in Proposition 2.5. (4) is a generalisation of the classical positivity: $\sum_{i,k}\langle v_i, \mathbb{E}[\rho_j(\exp((\xi_i - \xi_k)\wedge X))]v_k\rangle = \mathbb{E}\big[\big\|\sum_i \rho_j(\exp(\xi_i \wedge X))^* v_i\big\|^2\big] \ge 0$. (5) is Plancherel applied to the joint distribution on $\mathbb{R}^3 \times SU(2)$. (6) is the rotational equivariance proved in Proposition 2.6 extended via ρ_j. □ □

8.3 Sufficiency: a graded analogue of Bochner

The converse—that conditions (1)-(6) of Proposition 8.1 together imply existence of a probability law μ realising $\{F^{(j)}\}$—is, to our knowledge, an open question. We formulate the natural conjecture and defer the proof to subsequent work.

Proposition 8.2 (Bochner-graded conjecture)**.** *Let* $\{F^{(j)}\}_{j\in\frac{1}{2}\mathbb{N}_0}$ *be a family satisfying conditions (1)-(6) of Proposition 8.1. Then there exists a unique probability law* μ *on* $\mathbb{R}^3$ *with* $F^{(j)} = \Phi_\mu^{(j)}$ *for all* j*, provided the graded Plancherel transform* $\sum_j (2j+1)\hat{F}^{(j)}$ *is the Fourier transform of a non-negative measure on* $\mathbb{R}^3$*.*

A possible route to the proof: lift the family $\{F^{(j)}\}$ to a function $f\colon \mathbb{R}^3 \times SU(2) \to \mathbb{C}$ via the Peter-Weyl synthesis $f(\xi, g) = \sum_j (2j+1)\mathrm{tr}\big(F^{(j)}(\xi)\rho_j(g)^*\big)$, verify f is a positive-definite kernel on the group $\mathbb{R}^3 \times SU(2)$ (with appropriate group structure), and apply the classical Bochner theorem on locally compact abelian groups. The technical obstruction is that the wedge embedding $\mathbb{R}^3 \to SU(2)$ via $\xi \mapsto \exp(\xi \wedge X)$ is not a group homomorphism unless X is held fixed, complicating the direct application.

The Bochner-graded characterisation is the natural foundation for extensions of the present construction: parametric families characterised by graded constraints, hypothesis tests of distributional structure expressed at specific levels of the hierarchy, and inverse problems in non-commutative harmonic analysis applied to time-series geometry.

9 Closing

9.1 Repositioning Cramér-Kolmogorov

The classical scalar Cramér-Kolmogorov theory is, by Theorem 4.1, the $j = 0$ projection of the Plancherel-graded family $\{\Phi_\mu^{(j)}\}$. The relation is one of *repositioning*, not of displacement. Cramér's theorem on characteristic functions, Lévy's continuity theorem, the Cramér-Wold reduction, and the inversion formulae all survive intact at the abelian level. What changes is the algebraic ground: the abelian Fourier theory on $\mathbb{R}^d$ is no longer the foundational object but a projection of a richer non-commutative theory on the spin group $\mathrm{Spin}(d)$.

For $d = 3$ specifically, the spin group $\mathrm{Spin}(3) \cong SU(2)$ is compact, simply connected, three-dimensional, and admits a complete representation theory by half-integer indices. The Plancherel formula is exact and the irreducible decomposition of $L^2(SU(2))$ is at hand. The construction of this paper is consequently fully explicit at $d = 3$.

9.2 Companion to Paper 29: two faces of the foundation

The present paper is the algebraic-probabilistic component of a two-paper foundation. The companion is Paper 29 (Vázquez Broquá 2026c), which develops the parallel geometric-differential component: Hamilton's 1847 hodograph and Gauss's 1827 Theorema Egregium are instantiated on the kinematic embedding $\mathbf{q}(t) = (y_t, \Delta y_t, \Delta^2 y_t)$ of an empirical-stochastic time series, yielding two intrinsic geometric invariants—the curvature of the bivector trajectory and the Gaussian curvature of the lagged scalar surface—that supply the geometric structure that the Cartesian timeline lacks.

The two papers share the same algebraic substrate ($\mathrm{Cl}(3,0)$) and the same conceptual move (replace abelian-implicit foundations by explicit non-abelian extensions). Their difference is in the *modality* of probability vs. geometry, but the foundational logic is parallel:

	Paper 28 (this paper)	**Paper 29 (companion)**
Classical theory	Cramér-Kolmogorov	Cartesian timestamps
Algebraic substrate	$\mathbb{R}^d$ abelian additive	$\mathbb{R}$ ordered (no geometry)
Implicit kernel	$i \in U(1)$, central	none
Non-abelian replacement	$\mathrm{Spin}(3) \cong SU(2)$	$\mathrm{Cl}(3,0)$ kinematic embedding
Hierarchy	Plancherel-graded by j	Frenet-Serret + Egregium
Recovered classical	$j = 0$ projection	timestamp via grade-0
Foundational claim	probability is non-abelian	time series have geometry

Together, the two papers articulate a foundation on which the rest of the GSD programme—the moment tensor (Paper 24), the spectral identity (Paper 02), the entropy decomposition (Paper 04), the graded transfer entropy (Paper 05), the Takens embedding result (Paper 07), the moment-tensor concentration (Paper 25), and the twenty further papers of the programme—rest as derived consequences.

9.3 The genealogical loop closes

The present construction is the modern realisation of two readings that Hamilton's 1843 paper *On Fluctuating Functions* anticipated and that Cramér's 1942 abelianisation foreclosed. The argument-of-probability of Hamilton 1844 is the multivector $\mathbf{t}_{\mathrm{GSD}}$ of (23); the fluctuation-kernel of Hamilton 1843 is $\exp(\xi \wedge X)$ valued in Spin(3). Both are non-abelian where Cramér's choice was abelian; both are operationally accessible in 2026 thanks to the Clifford algebra apparatus that postdates Hamilton by over a century. The loop that opens with Hamilton 1843 closes here, with Cramér-Kolmogorov recovered as projection and the original non-abelian generality restored.

9.4 Open programme

We close with three open directions:

1. *Extension to $\mathbb{R}^d$ for $d > 3$.* The construction generalises naturally via $\mathrm{Cl}(d,0)$ with characteristic function families indexed by irreducible representations of $\mathrm{Spin}(d)$. The case $d = 4$ is particularly interesting: $\mathrm{Spin}(4) \cong SU(2) \times SU(2)$ admits a product hierarchy, and the Egregium-style invariants of the kinematic embedding for vector-valued time series are immediately accessible.

2. *Bochner-graded sufficiency.* Proposition 8.2 should be reduced to a proven theorem, which would close the algebraic foundation by giving a complete characterisation of which families $\{F^{(j)}\}$ are Plancherel-graded characteristic functions of probability laws.

3. *Statistical theory.* The statistical inference questions associated with the framework—hypothesis testing on $\Phi_\mu^{(j)}$, parameter estimation under the Plancherel metric, asymptotic distribution theory of empirical $\hat{\Phi}_\mu^{(j)}$—are open and constitute a substantial research programme in their own right. The empirical results of § 6 are a first probe; a complete theory remains to be built.

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